

What is deep learning?

- Deep learning is a subfield of Artificial Intelligence & Machine learning that is inspired by the structure of human brain.
- Deep learning algorithms attempt to draw similar conclusions as humans would by continually analyzing data with a given logical structure called neural network.

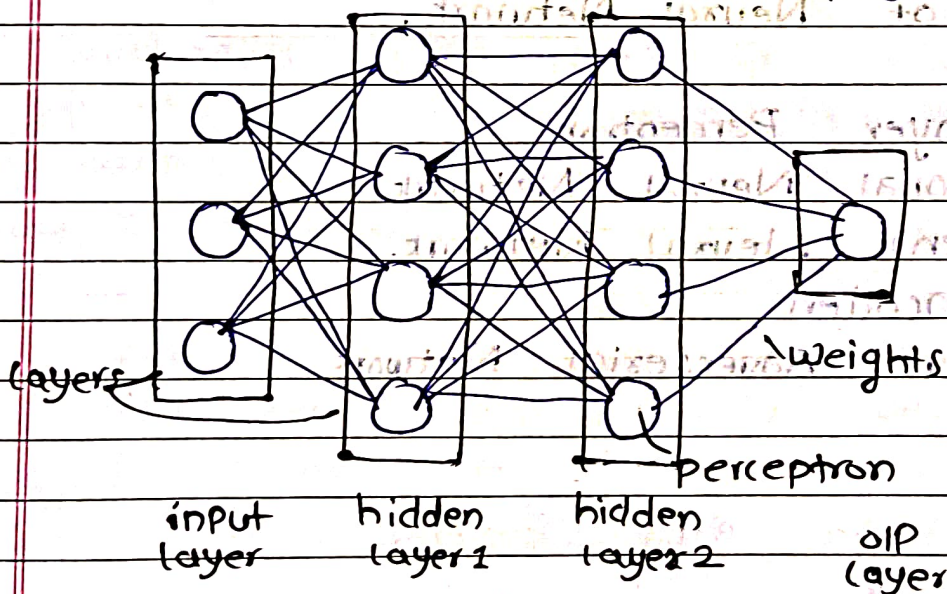
Artificial Intelligence

Machine Learning

Deep Learning

depends upon logical structure called Neural Network.

focuses on statistical techniques



ANN

Types of NN

• ANN

• CNN

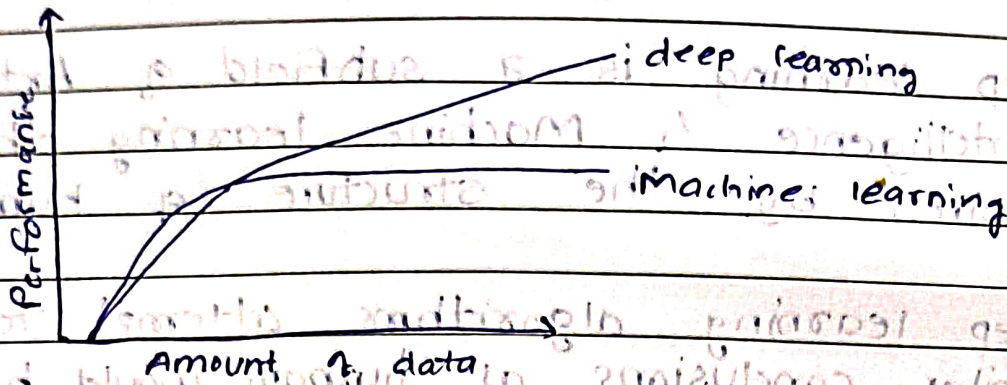
• RNN

• GAN

Representation Learning - DL model extracts features on their own.

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* DL VS ML



less data $\rightarrow$ ML wins
Large data $\rightarrow$ DL wins
DL is data hungry.

- ① Data dependency
- ② Hardware dependency
- ③ Training Time
- ④ Feature selection
- ⑤ Interpretability

* Types of Neural Network

- ① Multi-layer Perceptron
- ② convolutional Neural Network
- ③ Recurrent Neural Network
- ④ Auto-Encoders
- ⑤ Generative Adversarial Network